Week 1 - Mobile Robot Development with ROS2 (Jazzy Jalisco LTS)

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Long-term support

Resources

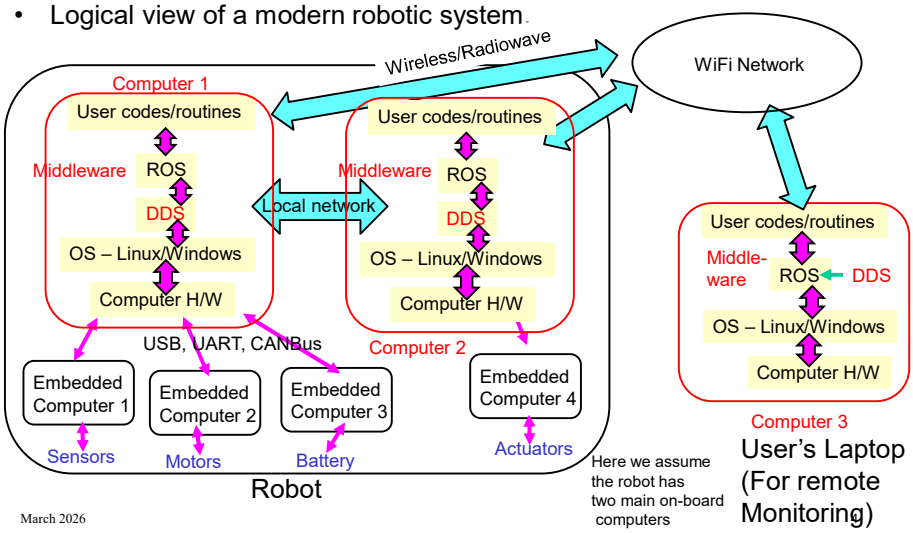
- Most documentations including getting started, tutorial, user manuals can be obtained from ROS2 official website https://ros.org and https://docs.ros.org/en/jazzy/index.html and github page.
- For learning ROS2, https://roboticsbackend.com/ and https://www.theconstructsim.com/ (Free)
- ROS2 makes use of DDS (distributed data systems) concepts. In particular it uses Open DDS standard (https://opendds.org/) to share data among various processes.
- In fact ROS2 is built on top of DDS/RTPS (Real-time publish subscribe protocol).

What is ROS2? (1)

- Although ROS stands for robot operating system, it is not an operating system like Unix, Linux, Windows or MacOS.
- In fact ROS2 is a collection of computer codes, or library that runs on top of a regular operating systems. It is a middleware that interface between your regular operating system and user codes.
- Currently only Linux and Window 10 are supported by ROS2.
- There is also ROS1, but it is gradually being replaced by ROS2 and will not be supported by 2026.
- ROS2 is based on a strongly-typed, anonymous publish/subscribe mechanism that allows for message passing between different processes.
- At the heart of any ROS2 system is the ROS graph. The ROS graph refers to the network of nodes in a ROS system and the connections between them by which they communicate.

What is ROS?(2)

- Logical view of a modern robotic system.



What is ROS2? (3)

- Just like our operating system having newer versions replacing older versions, ROS2 also has many versions, each version is called a *Distro* (short for distribution) is supported for 2 years and a certain version designated for long-term supported (LTS) will be supported for 5 years. Distros for plural
- The latest ROS2 LTS version is **Jazzy Jalisco**, which will be supported until 2029.
- You can find a comprehensive explanation and tutorials on using ROS2 on the ROS2 documentation site.

What is ROS2? (3)

- Current ROS2 distros (https://docs.ros.org/en/rolling/Releases.html):

Currently supported distros

List of Distributions

Below is a list of current and historic ROS 2 distributions. Rows in the table marked in blue are the currently supported distributions.

Distro	Release date	Logo	EOL date	ROS Boss
Kilted Kaiju	May 23, 2025		December 2026	Scott K Logan
Jazzy Jalisco	May 23, 2024		May 2029	Marco A. Gutiérrez
Iron Irwini	May 23, 2023		December 4, 2024	Yadunund Vijay
Humble Hawksbill	May 23, 2022		May 2027	Christophe Bédard / Audrow Nash
Galactic Geochelone	May 23, 2021		December 9, 2022	Scott Logan
Foxy Fitzroy	June 5, 2020		June 20, 2023	Jacob Perron / Dharini Dutia

Installation of Virtual Machine and Ubuntu OS

Some Notes on Installation of Oracle VM VirtualBox on Windows Laptop (1)

- Using **Oracle VM VirtualBox** running on **Windows 11** OS.
- **VirtualBox** is a virtual machine (VM) manager, which allows us to run other operating systems on top of the original operating system on our computer.
- You can download **VirtualBox** from https://www.oracle.com/my/virtualization/technologies/vm/downloads/virtualbox-downloads.html or virtualbox.org
- I installed VirtualBox on March 2025, so using version 7.1.6.
- **Ubuntu 24.04.3 LTS** will be installed inside VirtualBox to run ROS2 packages.
- For best performance (and not to run into hard disk memory issue later), I recommend allocate at least 3 CPU cores for Ubuntu, 4 gigabytes of RAM and 50 gigabytes of hard disk space. Personally, I allocate 4 CPU cores for Ubuntu, 8 GB of RAM and 100 GB hard disk space.

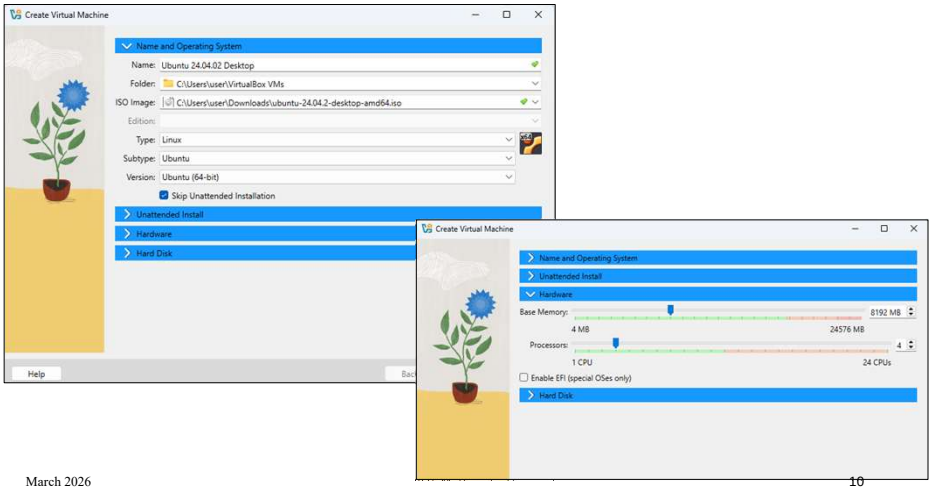
Some Notes on Installation of Oracle VM VirtualBox on Windows Laptop (2)

- First, we download the image for desktop Ubuntu 24.04.x (Noble Numbat) LTS, x is 1,2,3…. Just google for this, it's a file with *.iso suffix.
- Then run the Virtualbox and click 'New' to add a new virtual machine.
- It is advisable to enable 'Skip unattended installation' and install manually so that we can specify all the hardware attributes.
- Other hints are under the 'Display' tab, maximizes the Video Memory, set Monitor Count to 1, disable 'Enable 3D Acceleration' to prevent issues with the display later.



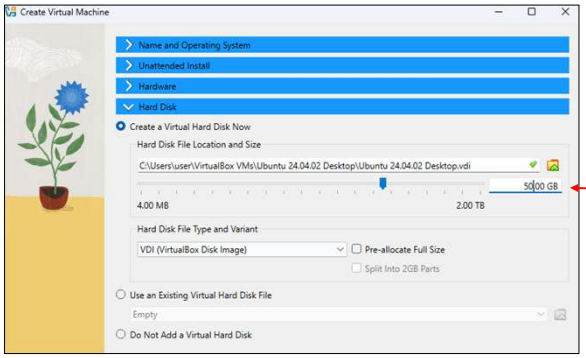
Installation Sequence of Linux on Virtual Machine (1)

- Set hardware property for virtual machine.



Installation Sequence of Linux on Virtual Machine (2)

- Set hard disk size for Ubuntu.



Recommended at least 40 GB hard disk space
Larger the better

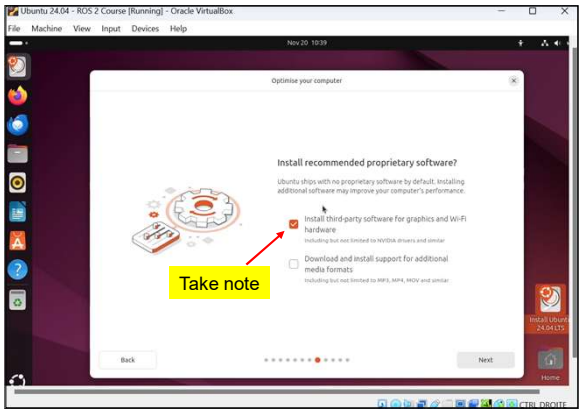
Installation Sequence of Linux on Virtual Machine (3)

In my case I encountered an issue after installing VirtualBox and trying to get it to run the Linux ISO file. The system will crash, reporting a certain driver could not be found and start. I use the following fix as documented in *StackOverflow*:

https://stackoverflow.com/questions/34778147/cannot-boot-windows-guest-in-virtualbox-without-kernel-module-error

Installation Sequence of Linux on Virtual Machine (4)

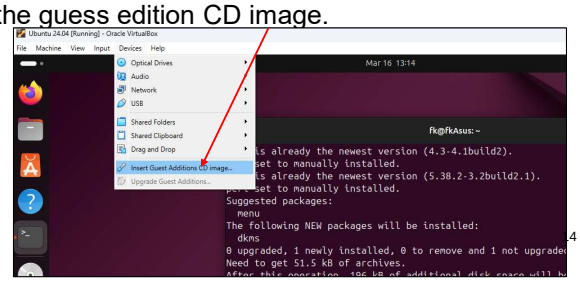
- Also, during the installation of Ubuntu make sure to check the '*Install recommended proprietary software*' checkbox, especially for computers with non-Intel graphic cards like Nvidia or Radeon.
- Else, we may faced issue while starting Ubuntu for the first time, which can be resolved by manually enabling the proprietary drivers.



Installation Sequence of Linux on Virtual Machine (5)

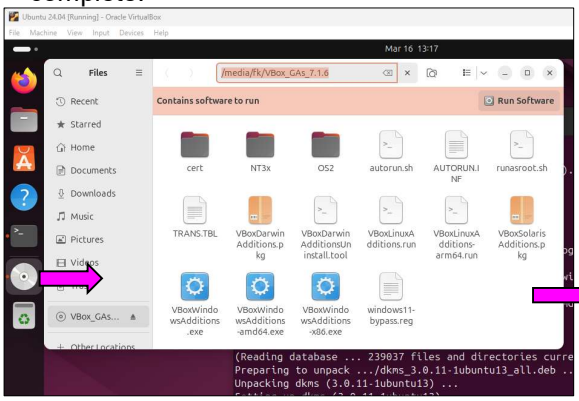
- Once Ubuntu is up-and-running, open a **Terminal**, and execute the commands `sudo apt update` followed by `sudo apt upgrade` to update the Ubuntu to the latest package.
- Another things to do is to install extra packages so that the Ubuntu window will resize correctly within the virtual machine frame.
- For this we install the following 5 packages:

  `Sudo apt install build-essential gcc make perl dkms`
- Once these are installed, we can go to the *menu->Devices* of the virtual machine and insert the guess edition CD image.



Installation Sequence of Linux on Virtual Machine (5)

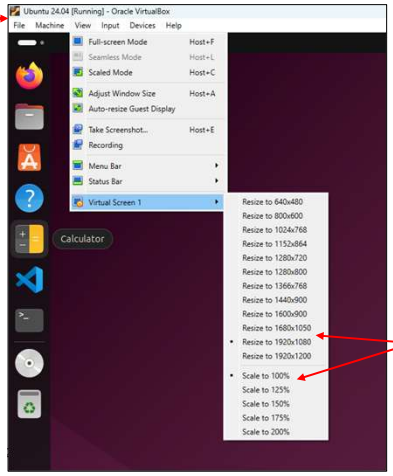
- A CD icon will appear on the left panel, right click this to open a file browser window. Then open the folder in a terminal and run the file *VBoxLinuxAddtions.run* (See instruction below).
- Now we can restart the Linux virtual machine and the setup will be complete.



Open in a Terminal and type
`sudo ./VBoxLinuxAdditions.run`
(use auto-completion)

Installation Sequence of Linux on Virtual Machine (6)

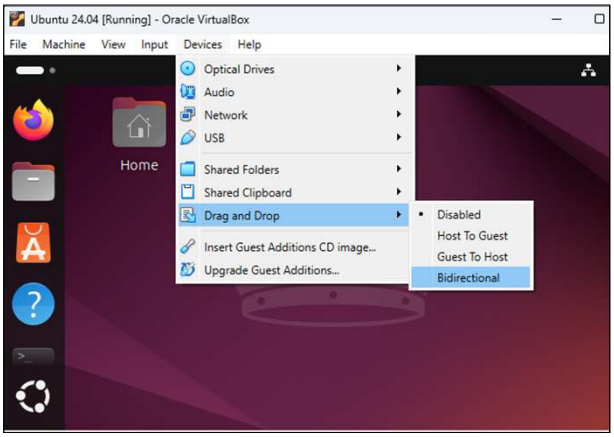
- Sometimes after this step the auto-resizing still doesn't work properly. In this case we can try manual resizing from the drop-down menu of the VirtualBox window.



Use this

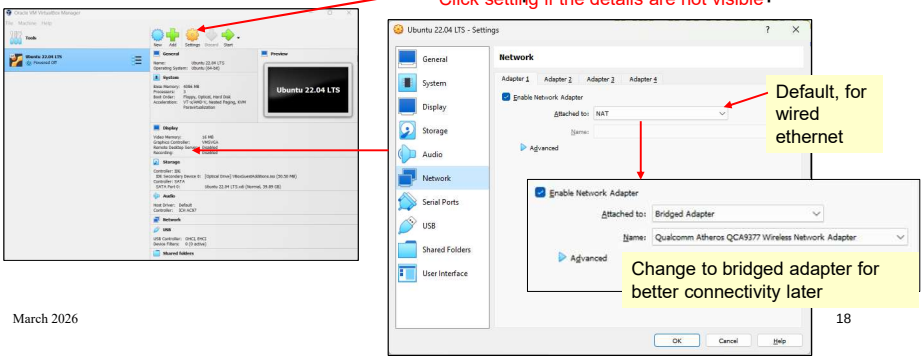
Installation Sequence of Linux on Virtual Machine (7)

- Finally, we want to enable cut/copy and paste between the host machine and the virtual machine. This can be done by enable bidirectional feature under ‘Drag-and-drop’ in the ‘Devices’ menu.



Some Notes on Installation of Oracle VM VirtualBox on Windows Laptop

- Make sure USB controller is enabled in VirtualBox Manager. Click the Network setting to select network access.
- For computer with wired ethernet: We can use the default “NAT” as shown.
- For using **WiFi** network, it is best to select the “Bridged Adapter” option and the name is the wireless network adapter card on the computer



Installation of ROS2 Jazzy

Summary of Steps to Install ROS2 Jazzy on Laptop with Virtual Machine (1)

- Download and install Oracle VM VirtualBox (Free version).
- Download Ubuntu Desktop 24.04 LTS ISO file ('iso' format is a disk image) from ubuntu.com.
- Create a new VM in VirtualBox, pointing to the ISO file downloaded. Follow instruction in create VM wizard to install Ubuntu 24.04 in VirtualBox. Allocate suitable RAM, hard disk space and CPU cores.
- Once Ubuntu is up-and-running, update Ubuntu by running:

```
sudo apt update
sudo apt upgrade
sudo apt autoremove
```

Summary of Steps to Install ROS2 Humble on Laptop with Virtual Machine (2)

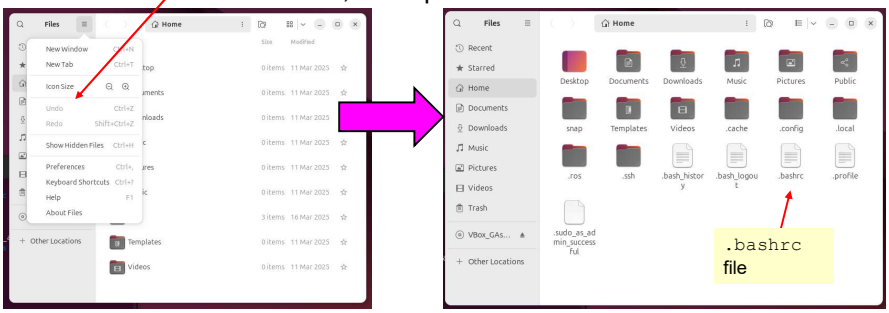
- Go to https://docs.ros.org/en/jazzy/Installation.html to install ROS2 Jazzy distro. Use the '*Debian packages*' option. Follow through all the instructions, including running an example ROS2 nodes at the end to verify that ROS2 Jazzy is properly installed.
- Finally install other extra tools, notably editor to edit your computer codes later. Here we recommend **VSCode**. In Linux, open Firefox web browser and google 'install VS code'.

Installation of ROS2 Jazzy (1)

- **ROS2 Jazzy** package is subsequently installed within the Ubuntu environment. The installation documentation on ROS webpage is quite complete, follow exactly. Here I am using the **Debian** packages option, the recommended, and install **ROS2 Jazzy** desktop version.
- After we complete the **ROS2 Jazzy** installation, we also need to install the development tools for building ROS packages, i.e. **colcon**. The installation documentation shows how to do that one shot, or we could do it manually.
- Other than the typical **Terminal** for Linux, for writing and editing the codes, I will be using **VSCode**. For **VSCode**, at least 3 extensions will be needed, C/C++, Python and CMAKE. We can also install the ROS extension from Microsoft.
- We can also use the improved terminal, called **Terminator** instead of the default Terminal app.

Edit *.bashrc* File in Home Directory (1)

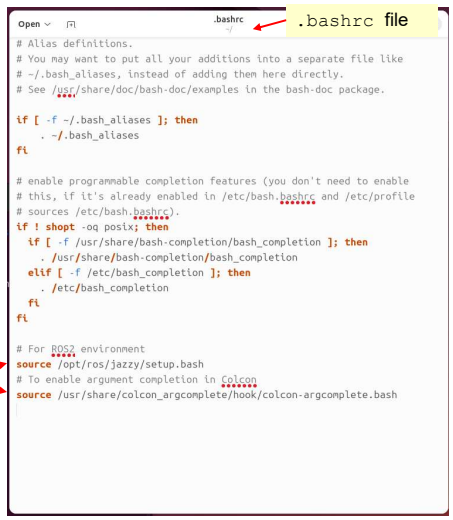
- Open the file browser app, it should point to your home directory.
- Enable the 'Show Hidden Files' option.
- Locate the `.bashrc` file, and open it in text editor.



Edit *.bashrc* File in Home Directory (2)

- Now we need to add ROS2 **Jazzy** installation folder in the search path of Linux so that the OS can find all the commands.
- It is good to enable the auto argument complement in **colcon** environment. We do this by appending the following source commands in the main `.bashrc` file in home directory.

These 2 source commands are added to .bashrc file in home directory.



Suggested Best Practices for Workspace

- One **workspace** for each project. A workspace is merely a folder to store all the project files, including compiled and executable files.
- Each unit of executable code is called a **package**. A package includes the function source codes, setup & configuration codes, installation instruction, compiled binaries and other executable files.
- A workspace can contain multiple packages.
- Also source the local `setup.bash` file of the workspace. This file is stored in `<workspace folder>/install`. This will be shown in the following example.